

Next-gen wireless communication through electronic-photonic integration

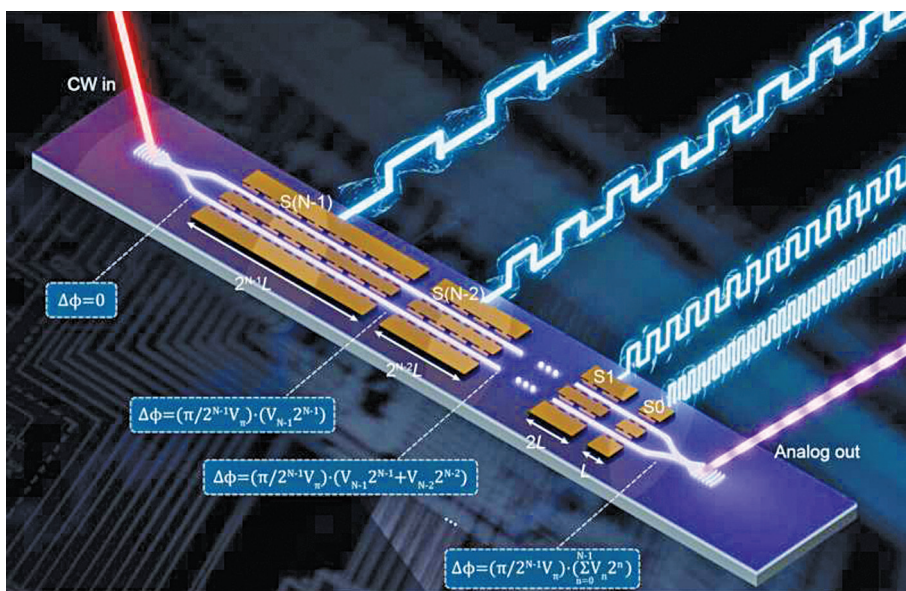


Illustration of the photonic-integrated electro-optic digital-to-analogue link (EO-DiAL) architecture (Credit: Lincar group/Harvard SEAS)

Harvard (Cambridge, Massachusetts) engineers have created a breakthrough device that directly converts digital electronic signals into analogue optical ones, bypassing bulky, power-hungry converters. Built on thin-film lithium niobate, the electro-optic digital-to-analogue link (EO-DiAL) compresses conversion into a single chip, delivering speeds up to 186 gigabits per second—nearly 100 times faster than home internet. Scalable and energy-efficient, it could revolutionise AI computing, data centres, fibre-optic networks, and next-gen wireless communications by redefining electronic-photonic integration.

Method found to prevent cracks in wearable electronics

Engineers at Brown University (Rhode Island) have uncovered a hidden weakness in flexible electronics, from wearables to foldable phones. They found that cracks in brittle ceramic electrode layers can spread into the polymer substrate, reducing durability and performance. By adding an intermediate layer to balance elastic mismatch, they prevented deep cracking across material combinations. This insight offers a roadmap for tougher, longer-lasting flexible devices, paving the way for more reliable wearables, solar panels, and foldables.

Substrate cracking in bent flexible PSCs (Credit: npj Flexible Electronics, 2025)

